

Problem Set 5

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1 Conjoint Experiments

Suppose you work for “Go Vote” (GV), a non-governmental organization interested in increasing voter turnout. During the primary election season, the organization will conduct an email campaign to estimate the treatment effect of several message components on voter turnout; it will then use the results in the general election.

The subject of the email will be one of “Vote Tuesday!”, “GV wants you to vote Tuesday!”, or “I personally want you to vote Tuesday!”. The email will either include or exclude a photo of GV’s Director. The email will close with a digital version of the Director’s signature including only her first name, her first and last names, or no signature. If the subject says “personally”, then a version of her signature must be included.

- a) Using `makeDesign()` in the `cjoint` package, create a conjoint design object for an experiment that will estimate the average marginal component effects (AMCEs) of the subject, photo, and signature. Show your code and print the contents of your design object. Use the example at `help(makeDesign)` for guidance.
- b) Consider the data from such a conjoint experiment, available on our GitHub page as `data/05-gv_conjoint.RData`, which you can import using the `load()` function. Estimate and interpret the average marginal component effects for each level of the three components. Provide a plot summarising the results.
- c) GV is also interested in the interaction between including the photograph and the signature. Estimate the average component interaction effect (ACIE) of the photograph and signature. Describe the evidence for or against the claim that the combination of photo and a signature together changes turnout *more* than does the photo and the signature alone.
- d) The conjoint experiment described above differs somewhat from the typical designs that we read about. Describe how it does so.